BIOcean5D — bacterial and Symbiodiniaceae responses to reciprocal transplantation between an impacted and a pristine bay

Results summary v3, 26 October 2026 · *Pocillopora meandrina* (POC) and *Porites rus* (POR) · Opunohu Bay (turbid, impacted) and Tiahura (pristine), Mo'orea. Supersedes v2 of the same date, which predates the heat-shock correction, the environmental-tracking polarity fix and the ITS2 analysis. Statistics from `Analysis/05-AAC-CommunityAnalyses.Rmd` (16S) and `Analysis/06-AAC-ITS2Symbionts.Rmd` (ITS2); full audit in `Manuscript/AAC-MethodsResults_Walkthrough.md`.

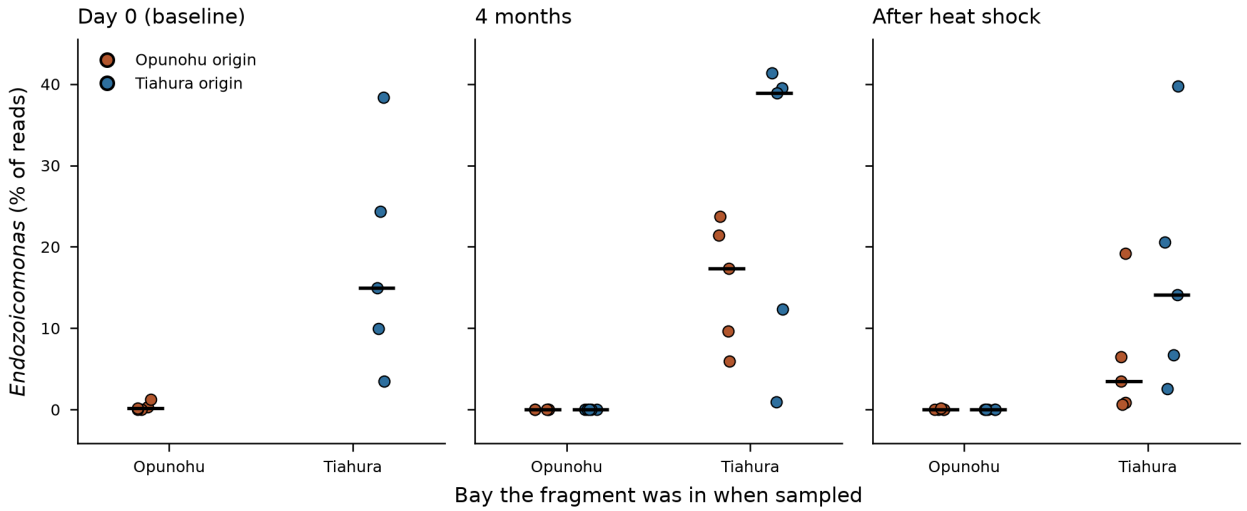
Methods

Colonies of both hosts (5 per bay per species) were reciprocally transplanted between Opunohu and Tiahura and sampled as mother-colony genotypes (day 0), at 2 weeks, at 4 months, immediately before an acute heat challenge and immediately after it, alongside free-living (0.2 µm) and particle-bound (3 µm) seawater and sediment references. Bacterial V4–V5 was amplified with 515F/926R, primers were removed with cutadapt, host reads depleted against the *Porites rus* assembly GCA_964035705.1 with bowtie2, and ASVs inferred with DADA2 from forward reads only (reverse quality was insufficient; `truncQ = 3`, `maxEE = 2`), with taxonomy from SILVA nr99 v138.2. Chimera removal left **1,795,938 reads (76.0% of input) in 15,067 ASVs**; after removal of chloroplast, mitochondrial and eukaryotic sequences, **14,075 prokaryotic ASVs** remained at a median depth of 10,214 reads. Sequences were aligned with MAFFT and a midpoint-rooted tree built with FastTree2. Symbiodiniaceae ITS2 amplicons from the same fragments (135 samples) were processed externally with SymPortal; because each sample carries essentially one ITS2 type profile (mean 1.1–1.4 detected), all symbiont statistics use the **post-MED sequence** layer — SymPortal's clade-C UniFrac distances and Bray–Curtis on post-MED sequence abundances. Community structure was tested by PERMANOVA (`vegan::adonis2`, 999 permutations) under a split-plot permutation design: whole colonies permuted freely for the between-colony term (bay of origin), samples permuted within colony for the within-colony terms (destination, timepoint). Dispersion homogeneity was checked with `betadisper`.

Design caveat. Sample identifiers encode the **bay of origin**, so a fragment's actual location is the opposite bay whenever it was out-transplanted — the distinction on which the interpretation turns. Losses fell complementarily: all 18 *Pocillopora* 2-week libraries failed at sequencing, and every *Porites* fragment at Tiahura died after 2 weeks, so all later *Porites* sat in Opunohu. Origin and destination are therefore fully crossed only in *Pocillopora* from 4 months onward, which is where the environmental and heat-shock contrasts are estimable.

Bacterial community

1. Destination, not origin, governs community composition, and *Endozoicomonas* is colonised from the surrounding water. In the balanced *Pocillopora* heat-shock window (n = 40, 10 colonies), the bay a fragment sat in explains **7.3% of Bray–Curtis** and 4.6% of Sørensen variation (p = 0.001 under within-colony permutation) and **8.05% of weighted UniFrac** (F = 3.19, p = 0.011); bay of origin, tested by permuting whole colonies, explains 4.6% (F = 1.83, p = 0.009). The taxon-level pattern is far stronger than any fitted model. At day 0 *Endozoicomonas* separates by provenance — Tiahura colonies 10/10 positive at a mean of 18.2% of reads, Opunohu 3/10 at 0.35%. By 4 months every Opunohu-origin colony moved to Tiahura had **gained** the genus (6.0–23.7%, from near zero) and every Tiahura-origin colony moved to Opunohu had **lost it entirely** (zero reads); within each origin, Mann–Whitney p = 0.0075 and Fisher on presence p = 0.0079. Across all timepoints, 35 of 35 fragments at Tiahura carry it against 2 of 33 at Opunohu.



***Endozoicomonas* is governed by destination, not origin.** Each point is a colony (fragments averaged within colony); bars are medians. Colour marks the bay of origin, position the bay the fragments were in when sampled. All 20 *Endozoicomonas* ASVs are included.

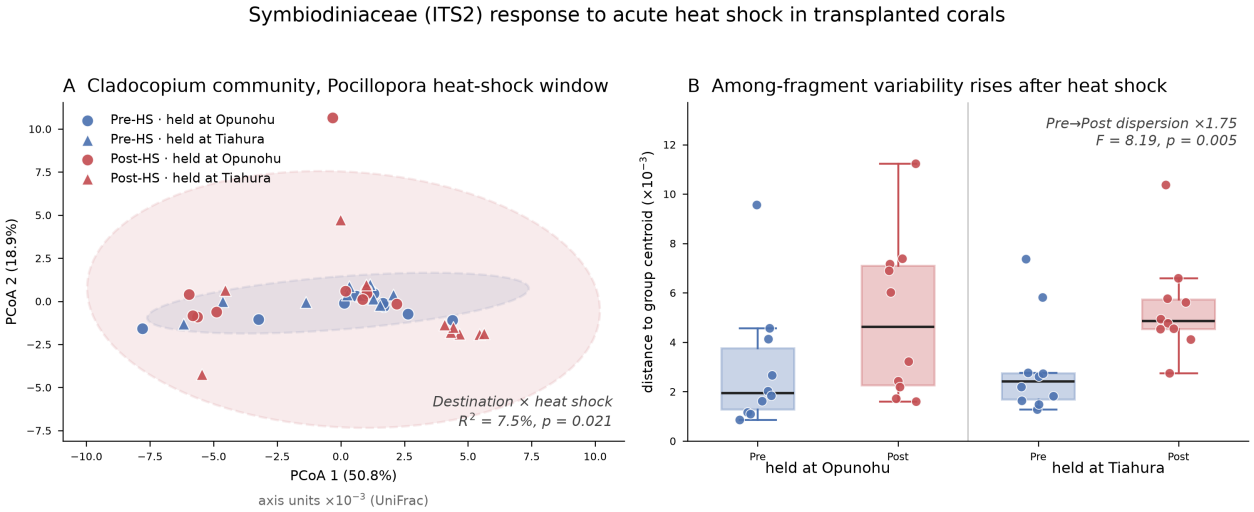
2. The acute heat shock is a tested null for bacteria. With the heat-shock subsets taken before the timepoint recode, the Pre-HS versus Post-HS term is non-significant in *Pocillopora* on every metric (Bray–Curtis R^2 = 0.026, p = 0.290; Sørensen

0.027, p = 0.221; weighted UniFrac 2.47%, p = 0.420) and the destination × heat-shock interaction is flat (p ≥ 0.712, weighted UniFrac p = 0.936). *Porites* likewise shows nothing (Bray–Curtis p = 0.125; weighted UniFrac R^2 = 5.24%, p = 0.734). Dispersion was homogeneous (destination F = 0.26, p = 0.609; timepoint F = 1.33, p = 0.267), so this is a genuine absence of response, not a power artefact of unequal spread. Communities are replaced rather than depleted over the experiment — total Sørensen dissimilarity 0.943 (POC) and 0.952 (POR), of which 98.0% and 98.5% is pure turnover — and no alpha-diversity measure responds to provenance or timeline in either host (all p ≥ 0.27).

Symbiodiniaceae community

3. Symbionts are host-controlled and site-insensitive — the mirror image of the bacteria. Host identity explains **72.5% of clade-C UniFrac variation** (p = 0.001; 65.9% on Bray–Curtis): *Pocillopora* carries C1/C42-type profiles, *Porites* the C15 lineage. Both hosts are essentially pure *Cladocopium* (97.3% and 98.9% of post-MED reads) and **Durusdinium is absent** (0.000% and 0.004%), so these corals hold no thermotolerant symbiont to shuffle towards. Unlike the bacteria, neither origin (p = 0.587) nor destination (p = 0.208) shifted the symbiont community of *Pocillopora* over four months in the opposite bay.

4. Heat shock destabilised the symbiont community instead of shifting it — an Anna Karenina response. The mean symbiont community did not move across the heat challenge (timepoint p = 0.568 UniFrac, 0.065 Bray–Curtis), but **among-fragment variability rose about 1.75-fold** in *Pocillopora* (UniFrac F = 8.19, p = 0.005; Bray–Curtis F = 8.61, p = 0.003). Variance rising while the mean holds still is the diagnostic signature of the Anna Karenina principle (Zaneveld, McMinds & Vega Thurber 2017), and it survives the obvious alternatives: the median dispersion ratio (2.30) exceeds the mean, so it is not outlier-driven; permutations blocked by colony give p = 0.003 and a paired test across the 20 colony × destination pairs that carry both timepoints gives 16 of 20 increasing (p = 0.0042); and although Post-HS libraries are shallower (median 39,213 vs 48,336 reads, p = 0.017), depth does not predict distance-to-centroid (ρ = –0.106, p = 0.515) and sequence richness is unchanged. The destination × heat-shock interaction is significant on both layers (R^2 = 7.5%, p = 0.021; 7.2%, p = 0.010) while neither main effect is, so the responses are idiosyncratic with a site-dependent direction rather than purely stochastic (Opunohu R^2 = 7.8%, p = 0.006; Tiahura R^2 = 11.2%, p = 0.043). *Porites* showed no such increase (ratio 1.24, p = 0.482). Critically, **the bacterial community of the same fragments did not destabilise** (betadisper on the heat-shock term F = 1.33, p = 0.267): the framework was formulated for the microbiome, yet here it is the obligate, host-selected symbiont that comes apart while the environmentally-acquired bacteriome holds. Taken with result 2, the two partners responded on orthogonal axes — bacteria tracked the environment and ignored the heat shock; symbionts ignored the environment and became erratic under it.



Symbiodiniaceae response to acute heat shock. (A) PCoA of clade-C UniFrac distances for the *Pocillopora* heat-shock window with 95% ellipses by timepoint; the Post-HS ellipse is visibly wider. (B) Distance to group centroid rises after heat shock in fragments held in either bay.

Limits and next steps

Differential abundance rests on DESeq2 alone (LEfSe and MaAsLin2 returned zero features in all four contrasts) and fold changes are unshrunken, so direction and prevalence are reportable but magnitudes are not; those tables still contrast origin and need refitting on destination with origin as a covariate and `lfcShrink` applied. The environmental-tracking analysis uses out-transplants only, so it cannot yet separate genuine tracking from coral microbiomes simply sitting nearer the Tiahura pool — the back-transplant controls needed are in the dataset. The *Porites* heat-shock window (n = 18) lies entirely at Opunohu and its null results are low-powered. SymPortal's clade-C phylogeny is taken as published. The Anna Karenina reading rests on treatment rather than measured stress — no bleaching score, photochemical yield or symbiont density has been brought into the analysis, and the "community" being destabilised is an assemblage of one to two closely related *Cladocopium* type profiles, not a diverse one. *Pocillopora* heat-shock RNAseq (34 libraries, single-end, duplicate runs to concatenate) remains to be quantified and will test whether the host transcriptional response aligns with the symbiont destabilisation reported here.